

Fostering antioxidant defences: up-regulation of antioxidant genes or antioxidant supplementation?

Vitamins have traditionally been considered as food components that are required in the normal diet to prevent deficiencies. However, a newer concept of the function of vitamins in nutrition has taken them beyond simply prevention of deficiency symptoms. This concept considers that many vitamins, when taken in relatively large doses, have important functions beyond preventing deficiencies. Linus Pauling was instrumental in putting forward this concept, particularly for vitamin C. Thus, relatively high intakes of vitamins, and in particular vitamins C and E which are antioxidants, are considered to be healthy for the human population. This may be true in some special situations such as, for instance, the prevention of Alzheimer's disease progression. However, recent epidemiological evidence has not supported the claim that antioxidant vitamins increase well-being and prolong life span. In fact, vitamin supplementation may be even detrimental and reduce life span. A new concept that we would like to put forward is that nutrients up-regulate the endogenous antioxidant defences. This is particularly true in the case of phytoestrogens for example, which bind to oestrogen receptors and eventually up-regulate the expression of antioxidant genes. In this review we discuss the pros and cons of antioxidant vitamin supplementation and also the possibility that the ingestion of some nutrients may be very effective in increasing antioxidant defences by up-regulating the activity of antioxidant enzymes which are normally present in the cell.